Exp No: 6 Time Series Analysis of Website Traffic

AIM:

To analyze the time series data of website traffic to identify the trends,seasonality and anomalies.

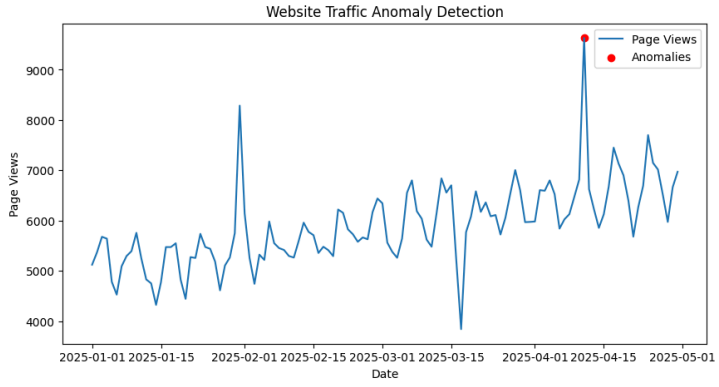
PROGRAM:

```
import pandas as pd
import numpy as np
np.random.seed(42)
dates = pd.date_range(start='2025-01-01', periods=120, freq='D')
page_views = (
    5000
    + np.arange(120) * 15
    + 500 * np.sin(2 * np.pi * np.arange(120) / 7)
    + np.random.normal(0, 250, 120)
)
page_views[30] += 2500
page_views[75] -= 2000
page_views[100] += 3000
data = pd.DataFrame({
    'Date': dates,
    'Page_Views': page_views.astype(int)
})
data.to_csv('website_traffic.csv', index=False)
print(data.head(10))
```

```
data = pd.read_csv(
    'website_traffic.csv',
    parse_dates=['Date'],
    index_col='Date'
)
from statsmodels.tsa.seasonal import seasonal_decompose
import matplotlib.pyplot as plt
data['Page_Views_Z'] = (
    (data['Page_Views'] - data['Page_Views'].mean())
    / data['Page_Views'].std()
)
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.figure(figsize=(10, 5))
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(
    anomalies.index,
    anomalies['Page_Views'],
    color='red',
    label='Anomalies'
)
plt.title('Website Traffic Anomaly Detection')
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.legend()
plt.show()
```

OUTPUT:

	Date	Page_Views
0	2025-01-01	5124
1	2025-01-02	5371
2	2025-01-03	5679
3	2025-01-04	5642
4	2025-01-05	4784
5	2025-01-06	4529
6	2025-01-07	5093
7	2025-01-08	5296
8	2025-01-09	5393
9	2025-01-10	5758



RESULT:

Thus the website traffic dataset is analyzed to identify trends,seasonality and anomalies.